Chapter 10

OSCILLATORS

CHAPTER OUTLINE

10-1 The Oscillator

10-2 Feedback Oscillator Principles

10-3 Sinusoidal Feedback Oscillators

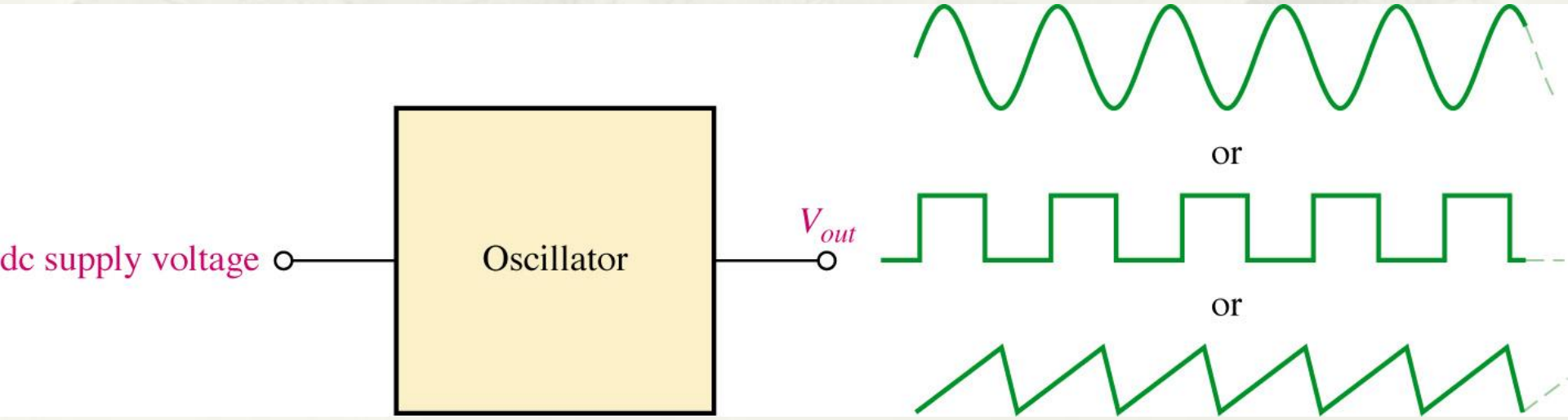
10-4 Relaxation Oscillators

10-1 The Oscillator

- **Oscillator**
  - An oscillator is a circuit that produces a periodic waveform on its output with only the dc supply voltage as a required input.
- **Oscillator Classification**
  - Feedback oscillator
  - Relaxation oscillator

Types of Oscillators

- **A basic oscillator**
  - All oscillators convert electrical energy from the dc power supply to periodic waveforms that can be used for various timing, control, or signal-generating applications.

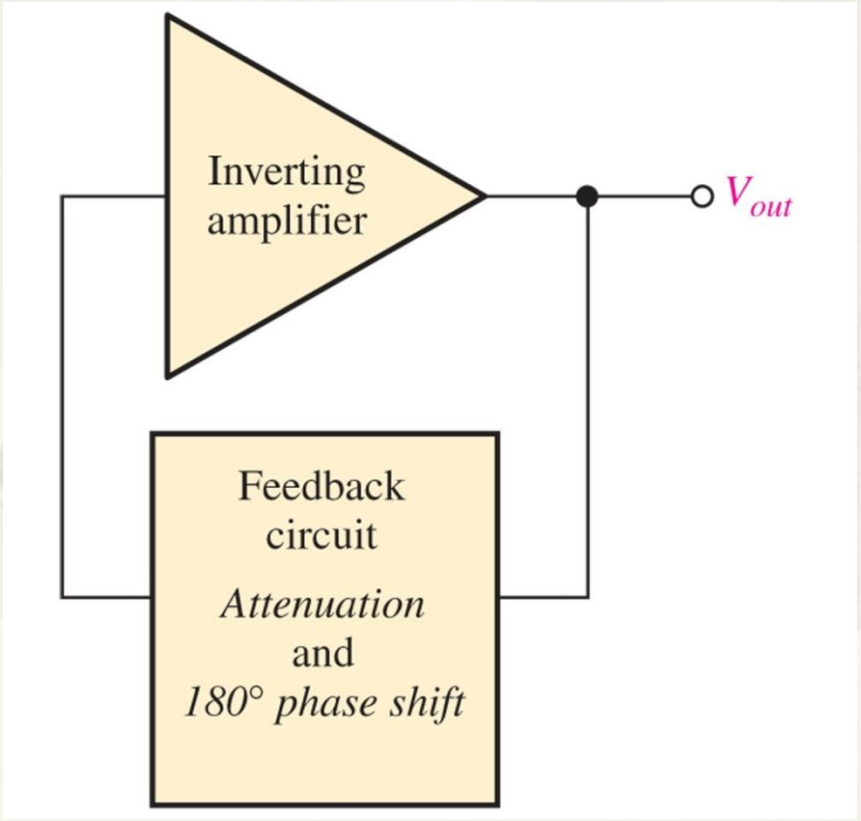


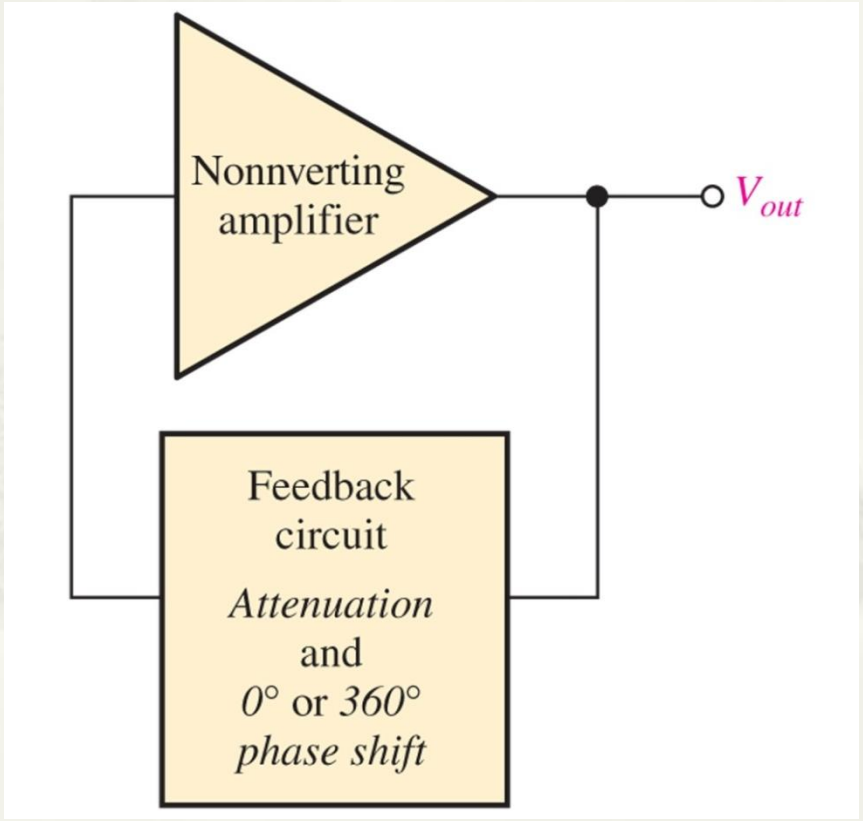
Types of Oscillators

- **Feedback Oscillators**
  - A feedback oscillator returns a fraction of the output signal to the input with no net phase shift, resulting in a reinforcement of the output signal.
  - A feedback oscillator consists of an amplifier for gain and a positive feedback circuit that produces phase shift and attenuates the signal.

Types of Oscillators

- **Feedback Oscillators**





There is no net phase shift around the loop.

Types of Oscillators

- **Relaxation Oscillators**
  - A relaxation oscillator uses an *RC* timing circuit instead of positive feedback to generate a waveform that is generally a square wave or other nonsinusoidal waveform.
  - Typically, a relaxation oscillator uses a Schmitt trigger or other device that changes states to alternately charge and discharge a capacitor through a resistor.

10-2 Feedback Oscillator Principles

- Positive Feedback
- Conditions for Oscillation
- Start-Up Conditions

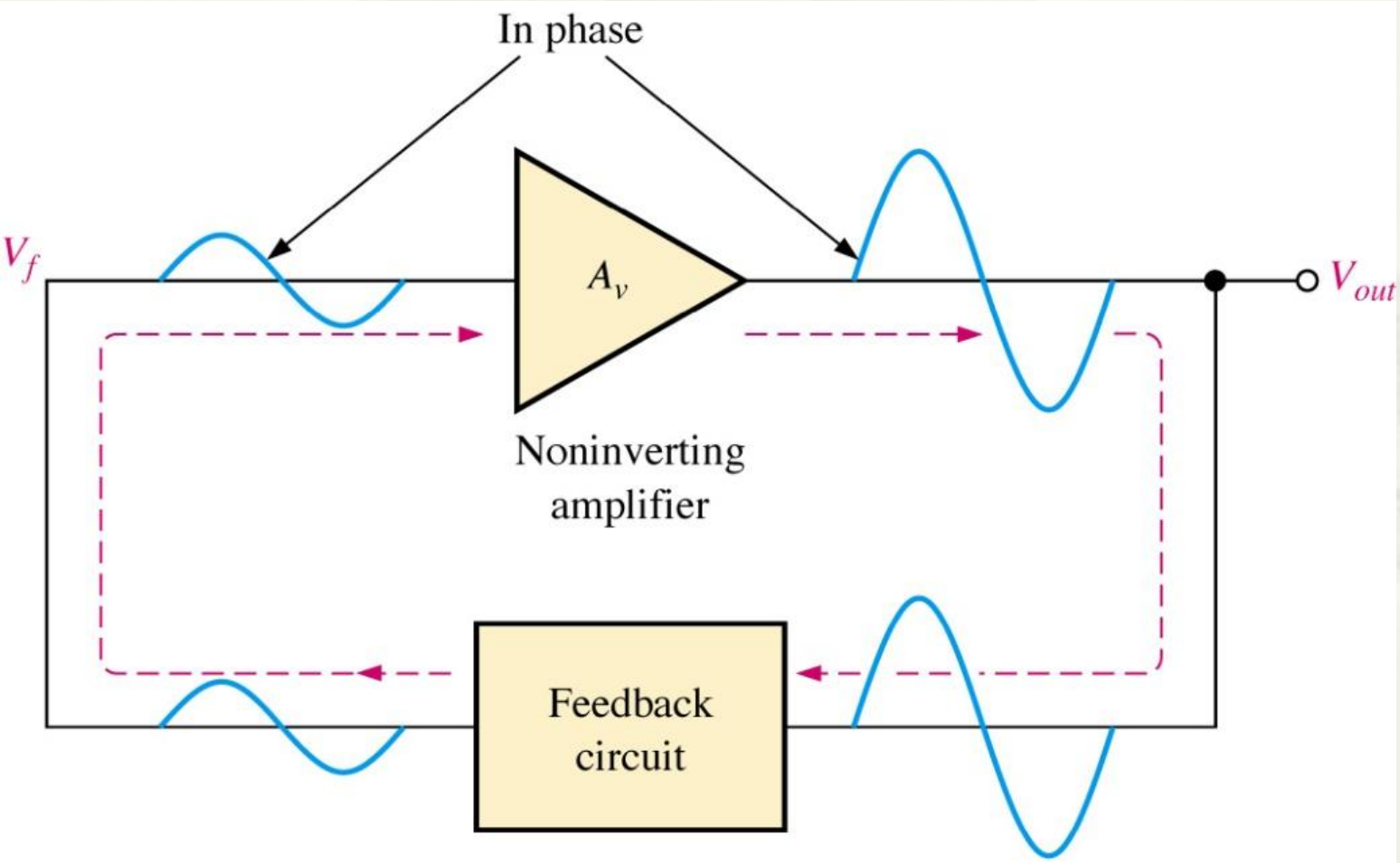
Feedback oscillators are used to generate sinusoidal waveforms.

Positive Feedback

- **Positive feedback**
  - Positive feedback is characterized by the condition wherein an in-phase portion of the output voltage of an amplifier is fed back to the input.
- **Oscillation**
  - A loop is create in which the signal sustains itself and a continuous sinusoidal output is produced. This phenomenon is called oscillation.

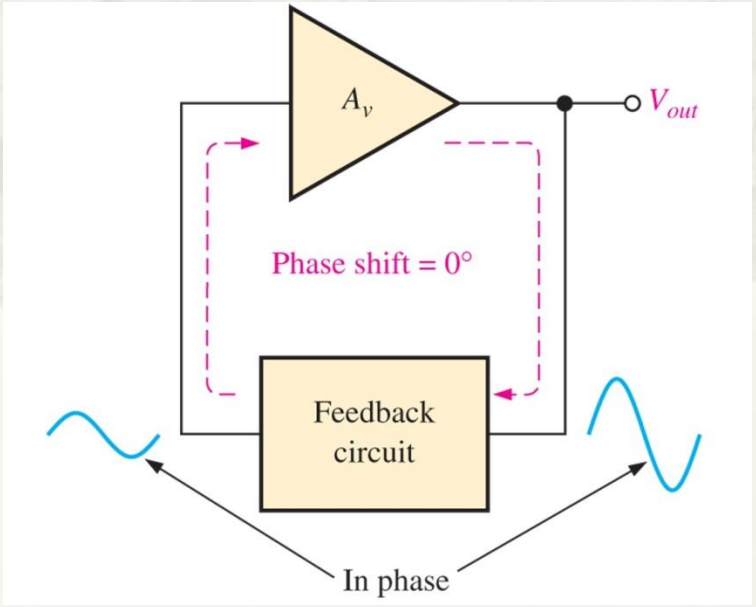
Positive Feedback

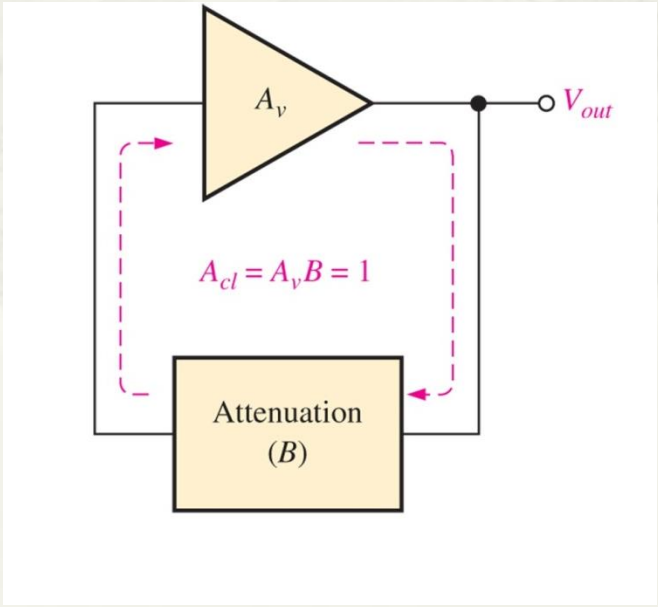
- **Positive feedback oscillation**



Conditions for Oscillation

- **Conditions required to sustain oscillations**
  - The phase shift around the feedback loop must be effectively 0° .
  - The voltage gain, A_{cl}, around the closed feedback loop (loop gain) must equal 1 (unity).





Conditions for Oscillation

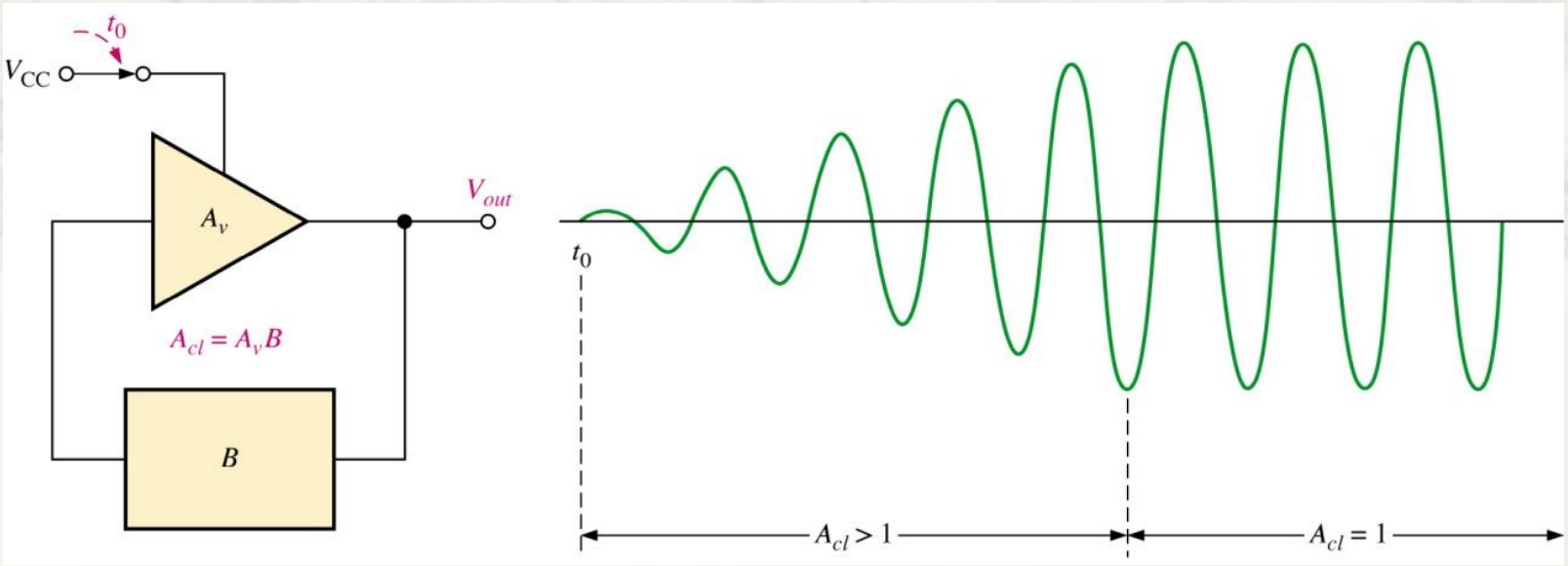
- **Conditions required to sustain oscillations**
  - The phase shift around the feedback loop must be effectively 0° .
  - The voltage gain, A_{cl}, around the closed feedback loop (loop gain) must equal 1 (unity).

If a sinusoidal wave is the desired output, a loop gain greater than 1 will rapidly cause the output to saturate at both peaks of the waveform, producing unacceptable distortion.

To avoid this, some form of gain control must be used to keep the loop gain at exactly 1, once oscillations have started.

Start-Up Conditions

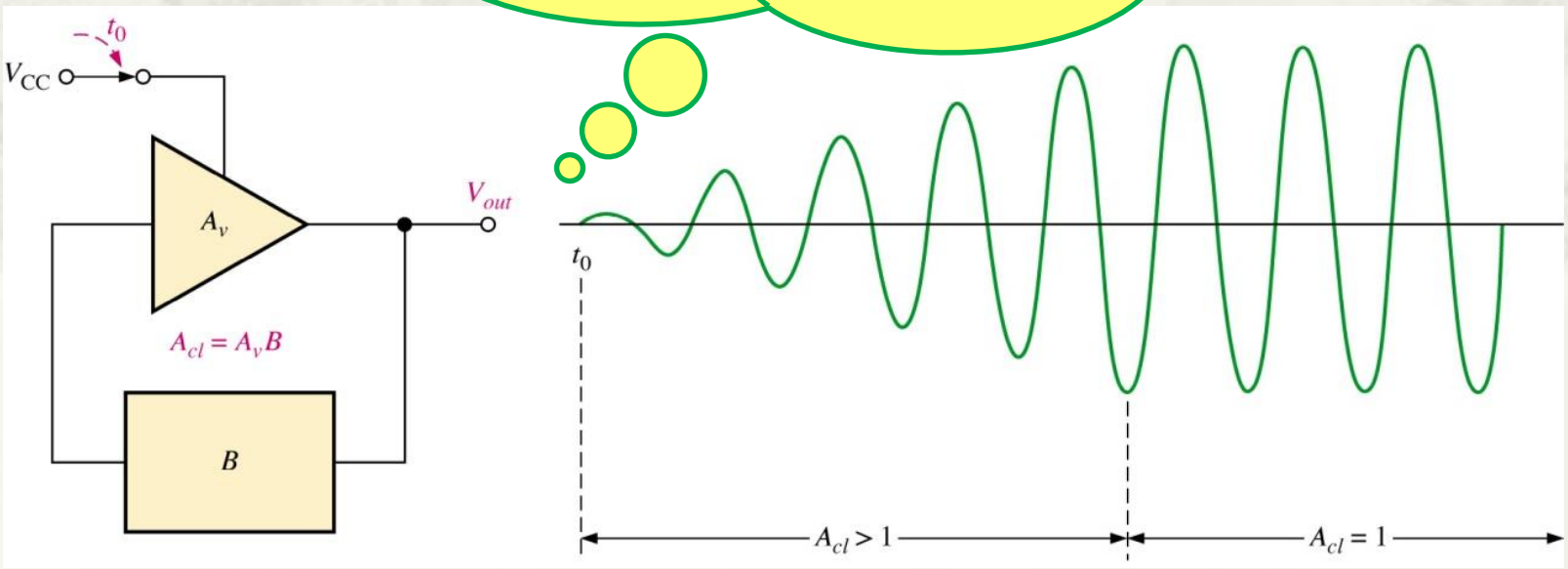
- **For oscillation to begin**
  - The voltage gain around the positive feedback loop must be greater than 1 so that the amplitude of the output can build up to a desired level.
  - The gain must then decrease to 1 so that the output stays at the desired level and oscillation is sustained.



Start-Up Conditions

- **For oscillation to begin**
  - The voltage gai… …ust be greater tha… …ld up to a…
  - The gain… …ays at the desired fr…

If the oscillator is initially off and there is no output voltage, how does a feedback signal originate to start the positive feedback?

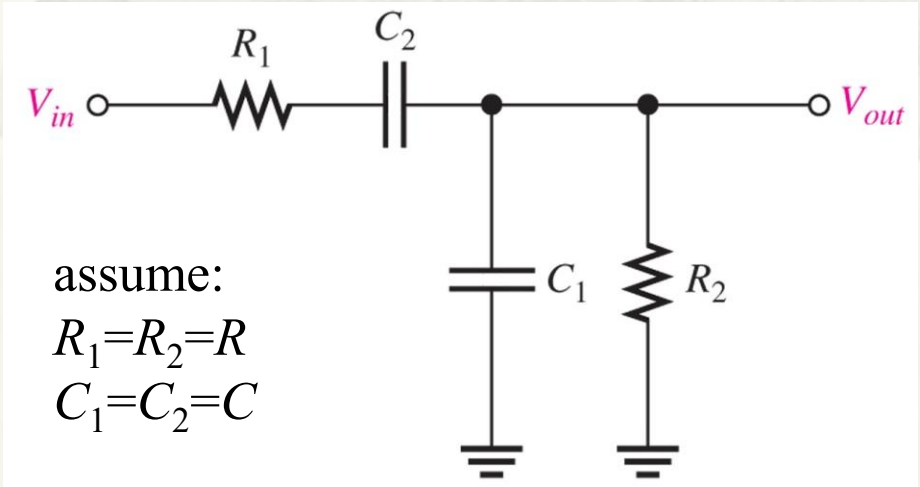


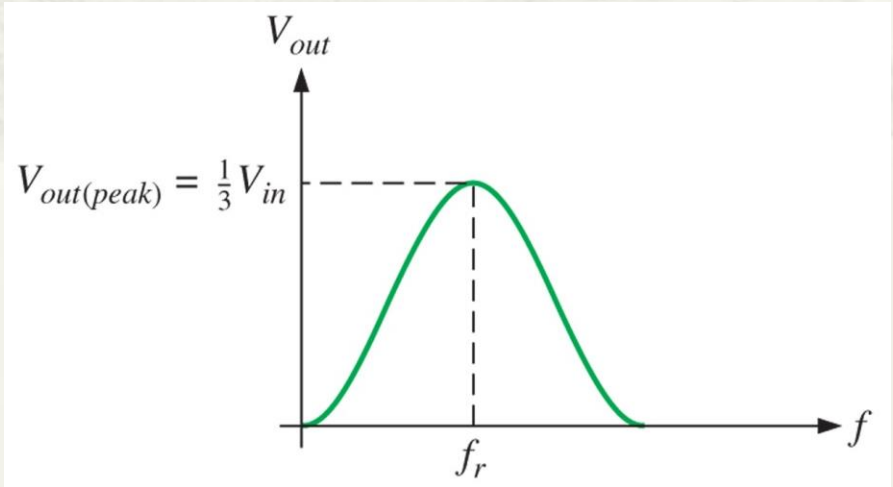
10-3 Sinusoidal Feedback Oscillators

- The Wien-Bridge Oscillator
- The Phase-Shift Oscillator
- Twin-T Oscillator

The Wien-Bridge Oscillator

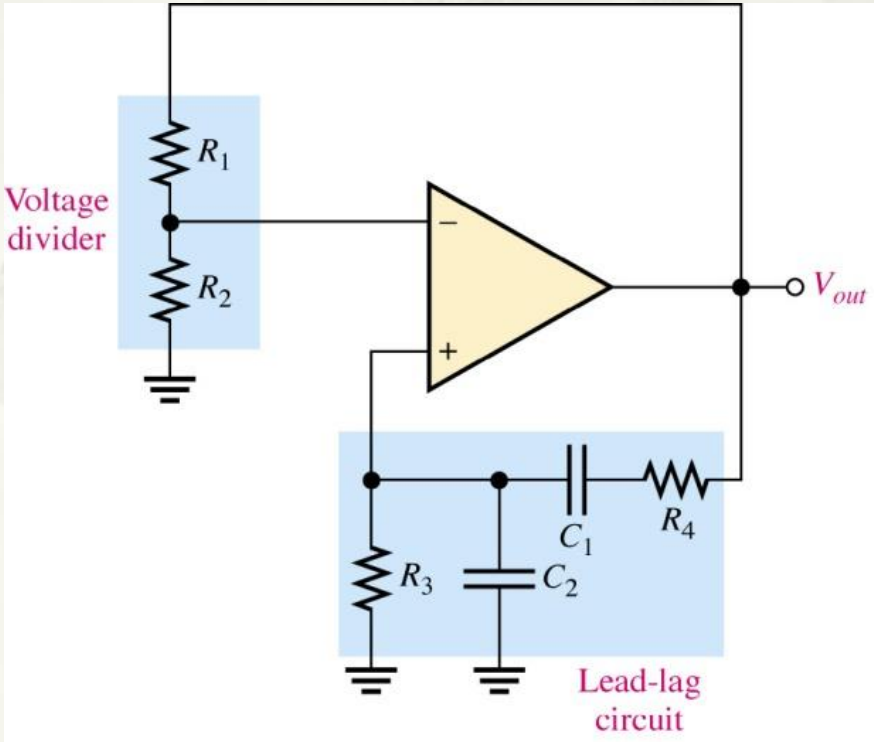
- **Lead-lag circuit**
  - The resonant frequency: $f_r = \frac{1}{2\pi RC}$
  - At the resonant frequency
    - ✓ $\frac{V_{out}}{V_{in}} = \frac{1}{3}$
    - ✓ Phase shift: 0°

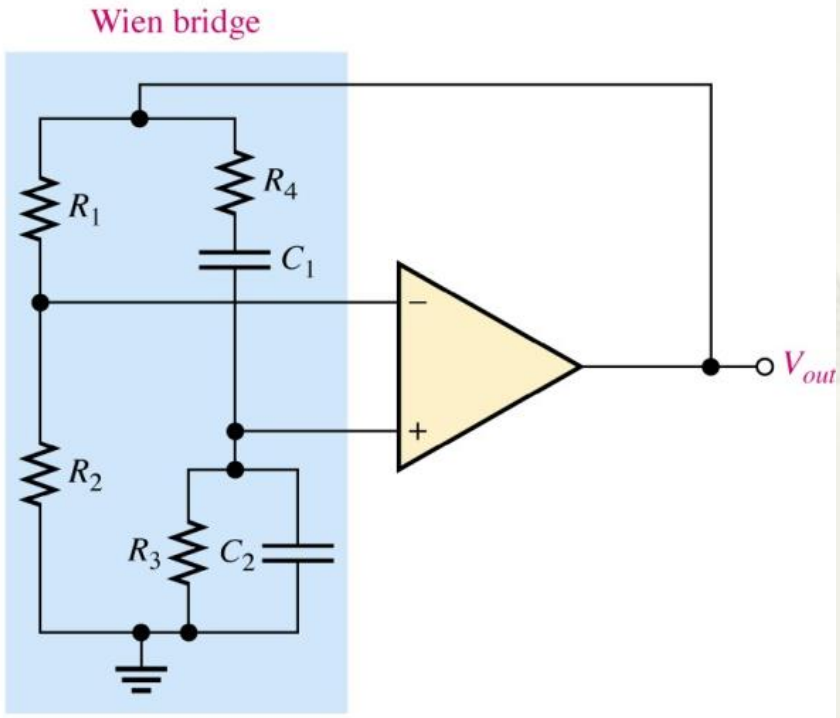




The Wien-Bridge Oscillator

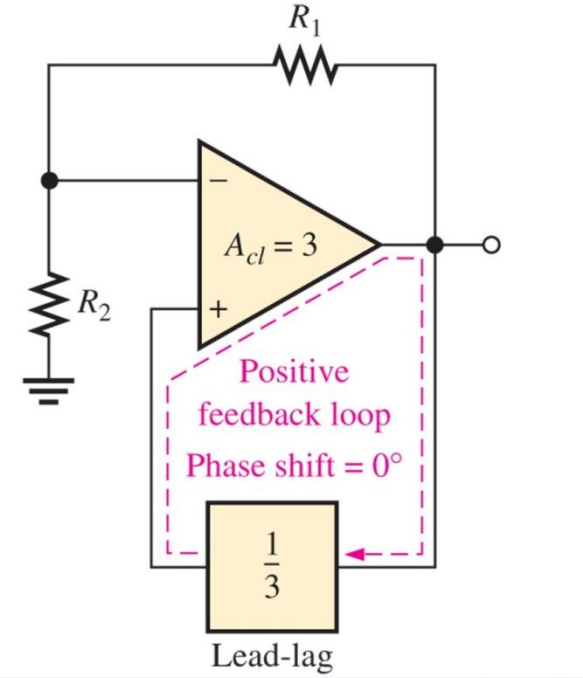
- **The Basic Circuit**
  - The Wien-bridge oscillator circuit can be viewed as a noninverting amplifier configuration with the input signal fed back from the output through the lead-lag circuit.

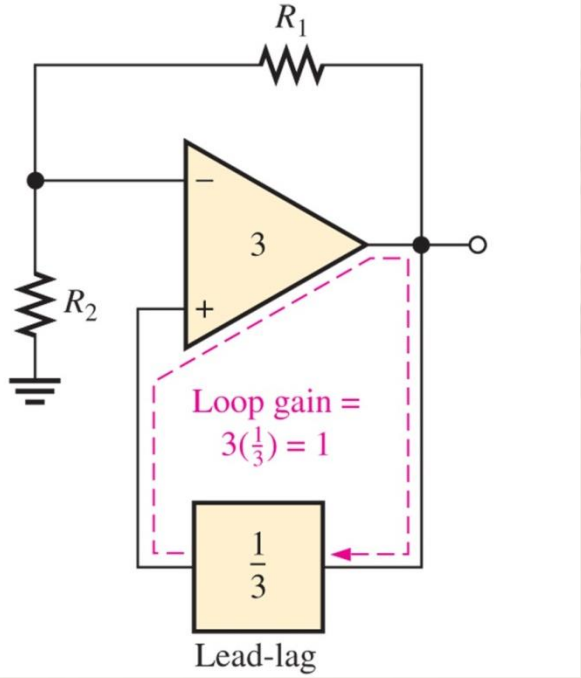




The Wien-Bridge Oscillator

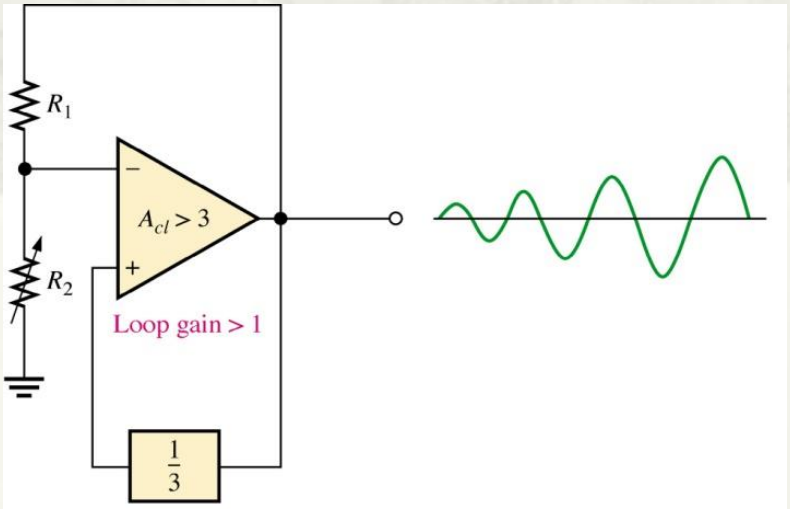
- **Positive Feedback Conditions for Oscillation**
  - The phase-shift condition: Phase shift = 0°
  - The unity-gain condition in the feedback loop: $A_{cl} = 3$
  - $R_1 = 2R_2, A_{cl} = \frac{R_1+R_2}{R_2} = 3$

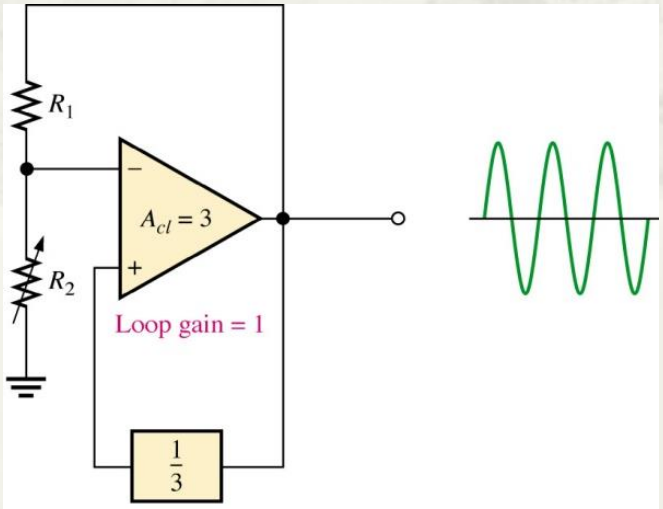




The Wien-Bridge Oscillator

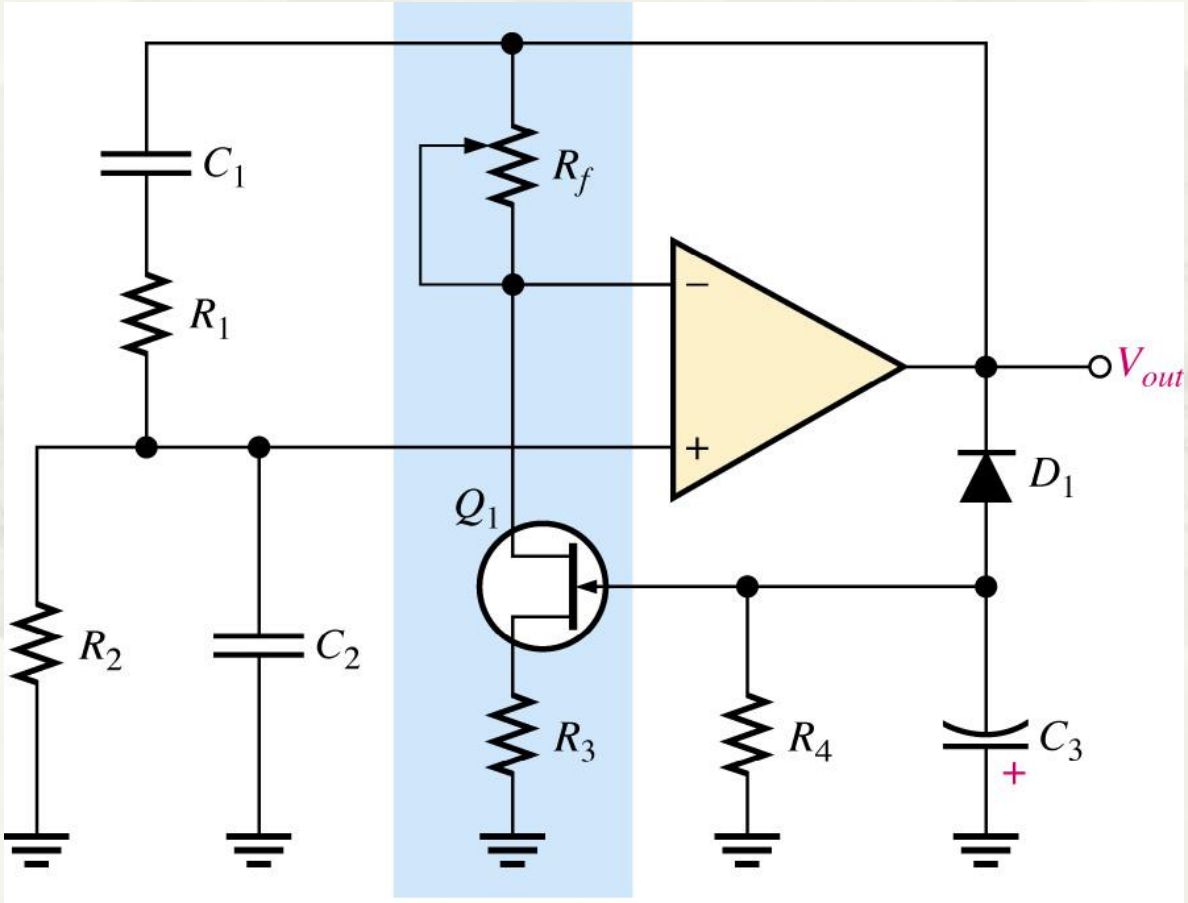
- **Start-Up Conditions**
  - Initially, the closed-loop gain of the amplifier itself must be more than 3 ($A_{cl} > 3$) until the output signal build up to a desired level.
  - Ideally, the gain of the amplifier must then decrease to 3 so that the total gain around the loop is 1 and the output signal stays at the desired level, thus sustain oscillation.





The Wien-Bridge Oscillator

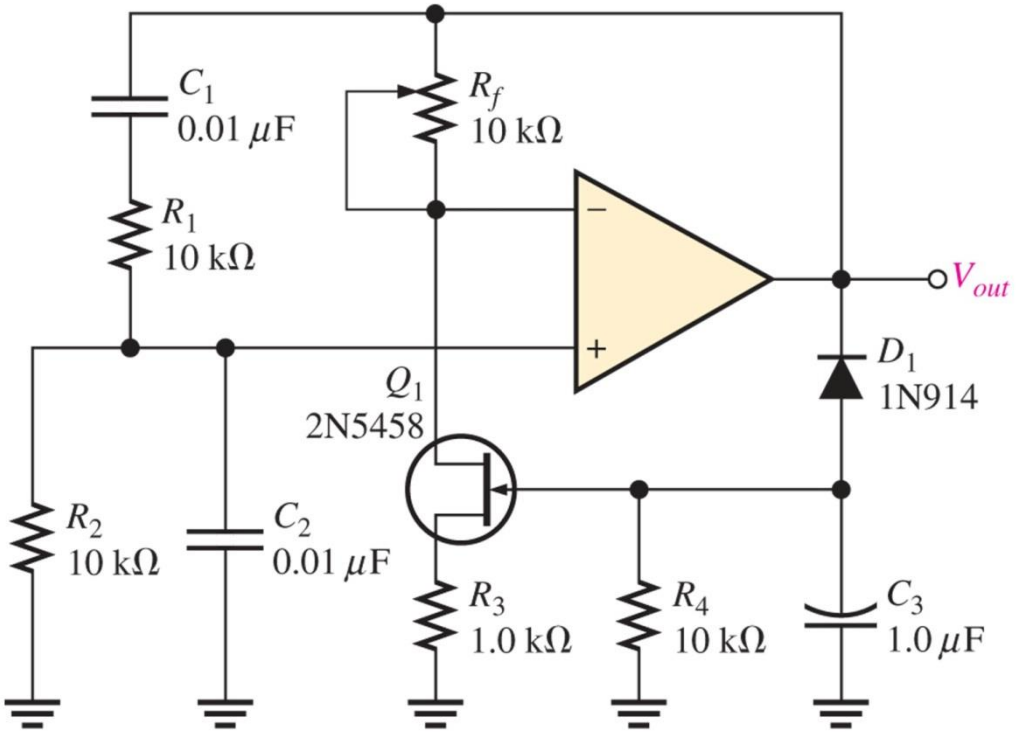
- **Self-starting Wien-bridge oscillator**
  - Use a JFET in the negative feedback loop



Example 10-1

Problem

Determine the frequency for the Wien-bridge oscillator in Figure 10–11. Also, calculate the setting for R_f assuming the internal drain-source resistance, r'_{ds}, of the JFET is 500 Ω when oscillations are stable.



Example 10-1

Solution

For the lead-lag circuit, $R_1 = R_2 = R = 10\,\text{k}\Omega$ and $C_1 = C_2 = C = 0.01\,\mu\text{F}$. The frequency is

$$f_r = \frac{1}{2\pi RC} = \frac{1}{2\pi(10\,\text{k}\Omega)(0.01\,\mu\text{F})} = \mathbf{1.59\ kHz}$$

The closed-loop gain must be 3.0 for oscillations to be sustained. For an inverting amplifier, the gain is that of a noninverting amplifier.

$$A_v = \frac{R_f}{R_i} + 1$$

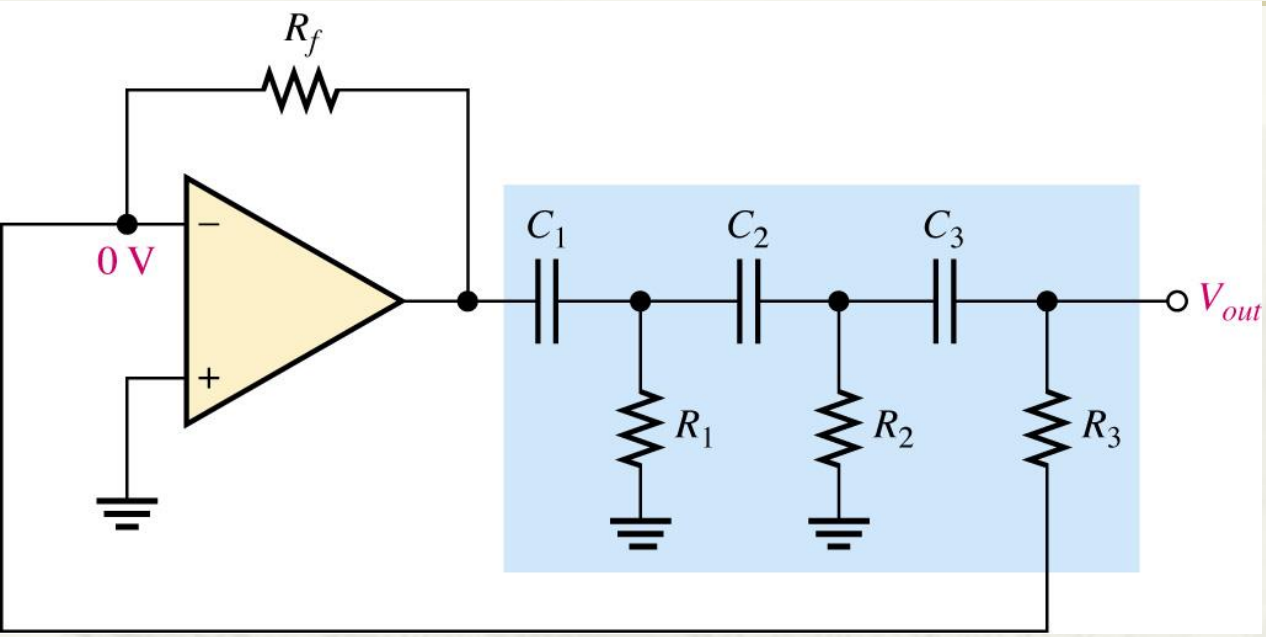
R_i is composed of R_3 (the source resistor) and r'_{ds}. Substituting,

$$A_v = \frac{R_f}{R_3 + r'_{ds}} + 1$$

Rearranging and solving for R_f,

$$R_f = (A_v - 1)(R_3 + r'_{ds}) = (3 - 1)(1.0\,\text{k}\Omega + 500\,\Omega) = \mathbf{3.0\ k\Omega}$$

The Phase-Shift Oscillator



Oscillation occurs at the frequency where the total phase shift through the three RC circuits is 180° .

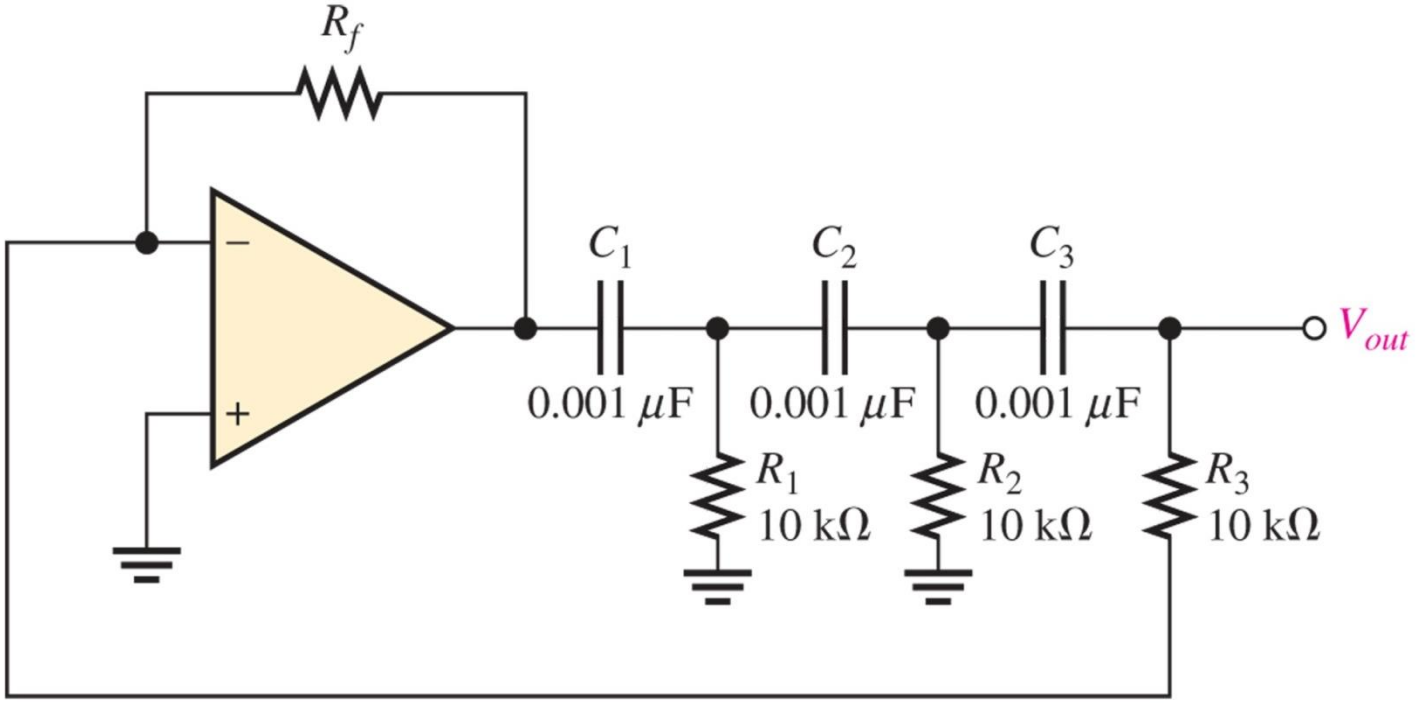
- **The frequency of oscillator**
  - $f_r = \frac{1}{2\pi\sqrt{6}RC}$, where $R_1 = R_2 = R_3 = R$ and $C_1 = C_2 = C_3 = C$
- **The attenuation of the three-section *RC* feedback circuit**
  - $B = \frac{1}{29}$, where $B = R_3/R_f$

Example 10-2

Problem

(a) Determine the value of R_f necessary for the circuit in Figure 10–13 to operate as an oscillator.

(b) Determine the frequency of oscillation.



Example 10-2

Solution

(a) $A_{cl} = 29$, and $B = \dfrac{1}{29} = \dfrac{R_3}{R_f}$.

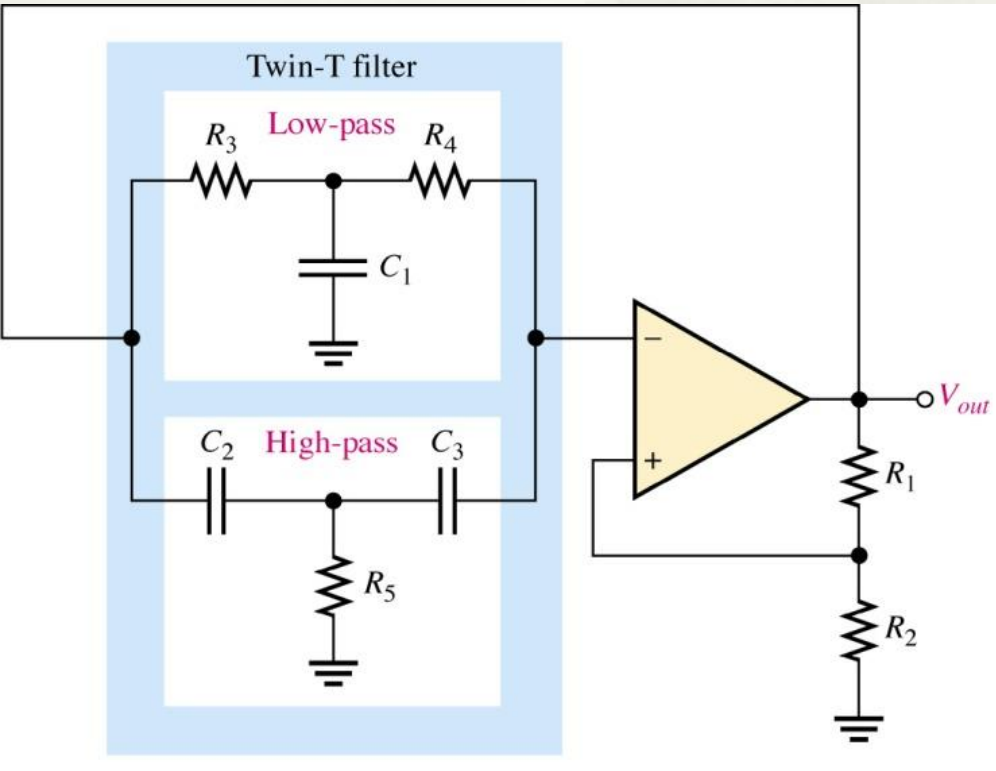
$$\frac{R_f}{R_3} = 29$$

$$R_f = 29R_3 = 29(10\,\text{k}\Omega) = \mathbf{290\ k\Omega}$$

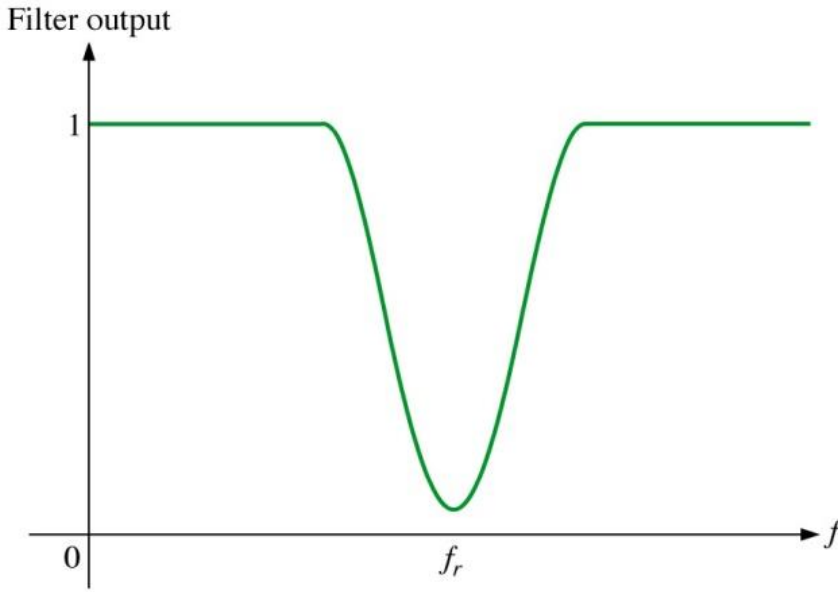
(b) $R_1 = R_2 = R_3 = R$ and $C_1 = C_2 = C_3 = C$.

$$f_r = \frac{1}{2\pi\sqrt{6}RC} = \frac{1}{2\pi\sqrt{6}(10\,\text{k}\Omega)(0.001\ \mu\text{F})} \cong \mathbf{6.5\ kHz}$$

Twin-T Oscillator



(a) Oscillator circuit



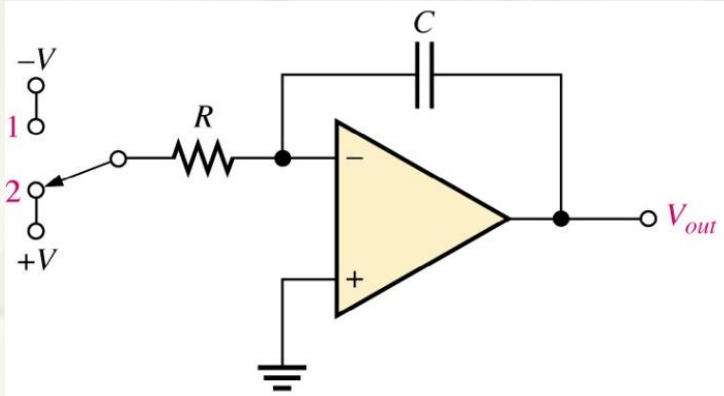
(b) Twin-T filter's frequency response curve

10-4 Relaxation Oscillators

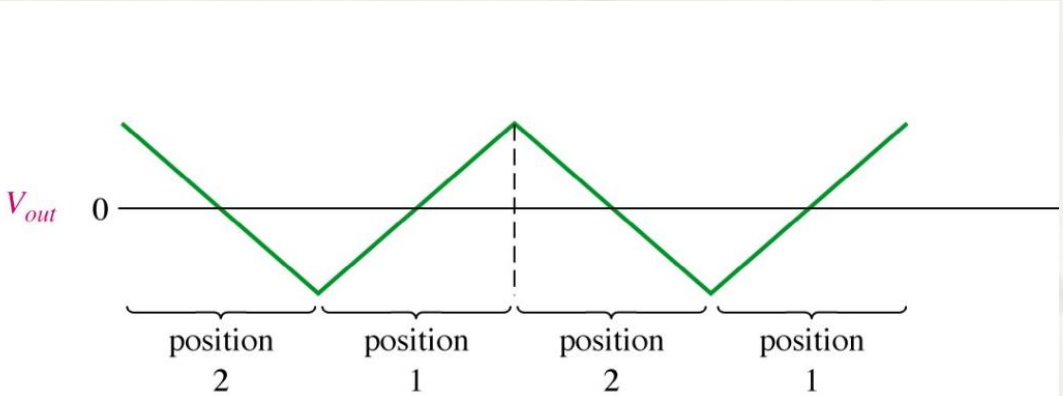
- A Triangular-Wave Oscillator
- A Square-Wave Oscillator

A Triangular-Wave Oscillator

- **The basic idea**
  - Use a dual-polarity, switched input and an op-amp integrator



(a)

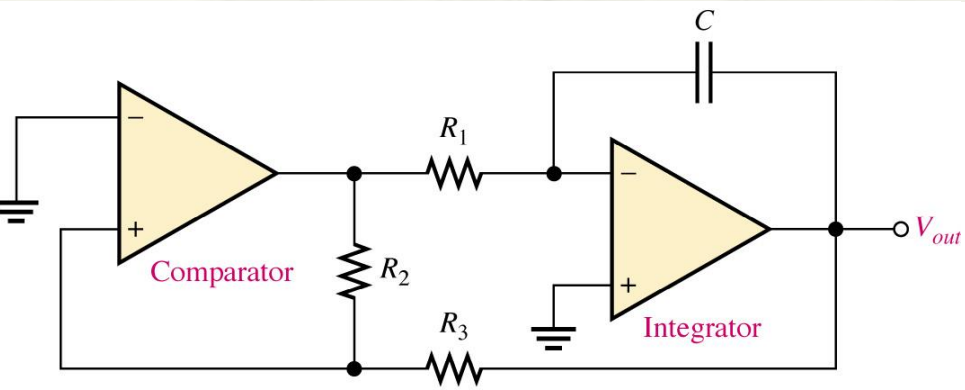


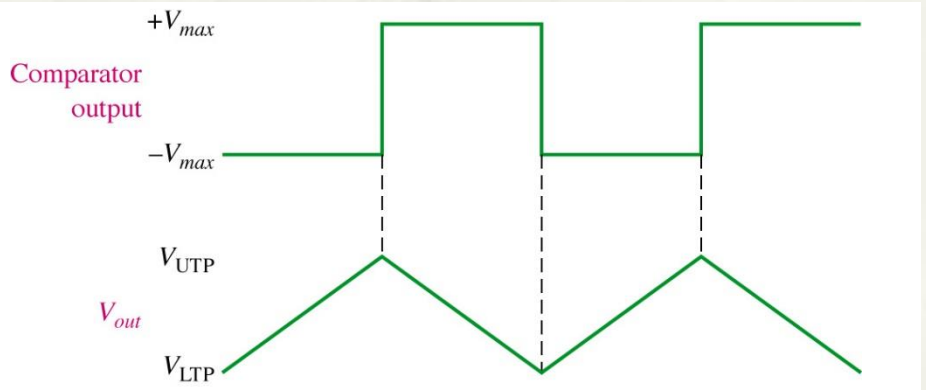
(b) Output voltage as the switch is thrown back and forth at regular intervals

A Triangular-Wave Oscillator

- **A Practical Triangular-Wave Oscillator**
  - Utilizes an op-amp comparator set up as a Schmitt trigger to perform the switching function.

Function Generator





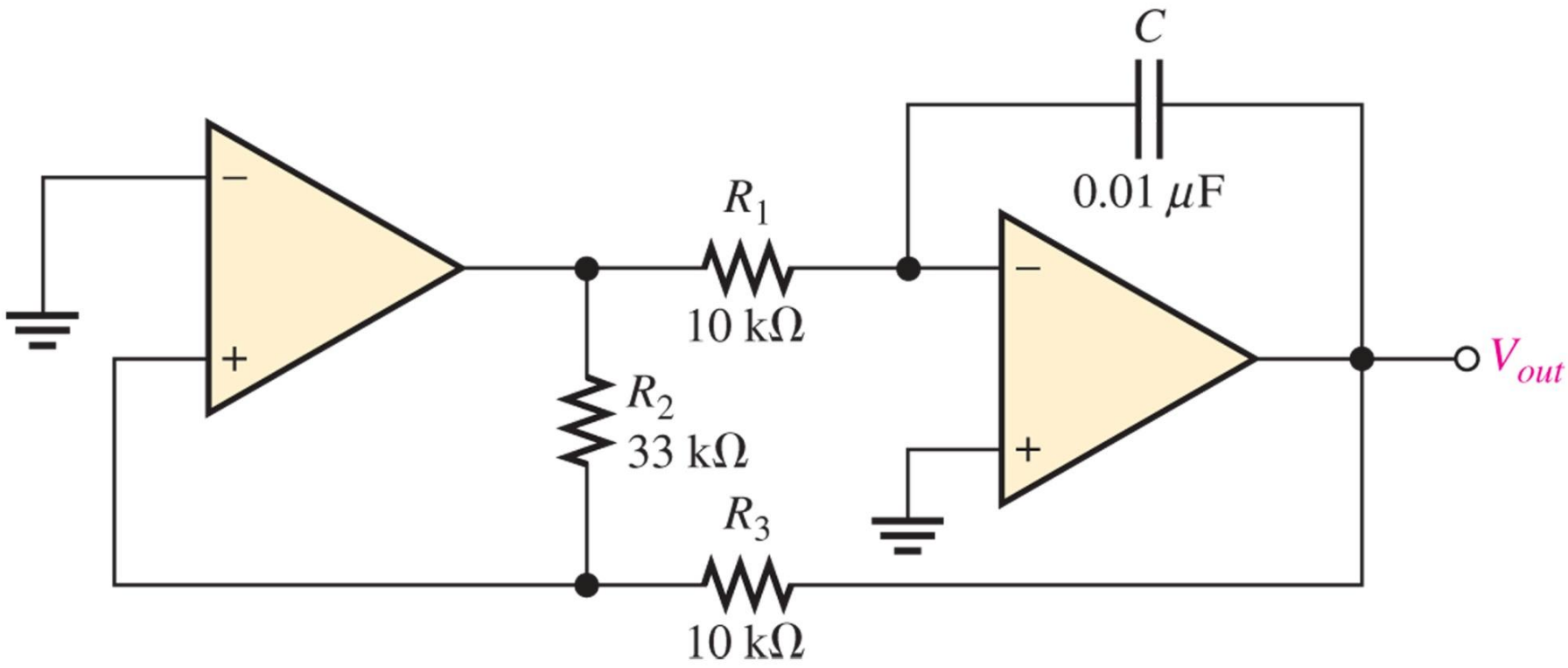
Both a triangular-wave generator and a square-wave generator

The approximate frequency of oscillator: $f \cong \frac{1}{4R_1C}\left(\frac{R_2}{R_3}\right)$

Example 10-3

Problem

Determine the approximate frequency of oscillation of the circuit in Figure 10–18. To what value must R_1 be changed to make the frequency 20 kHz?



Example 10-3

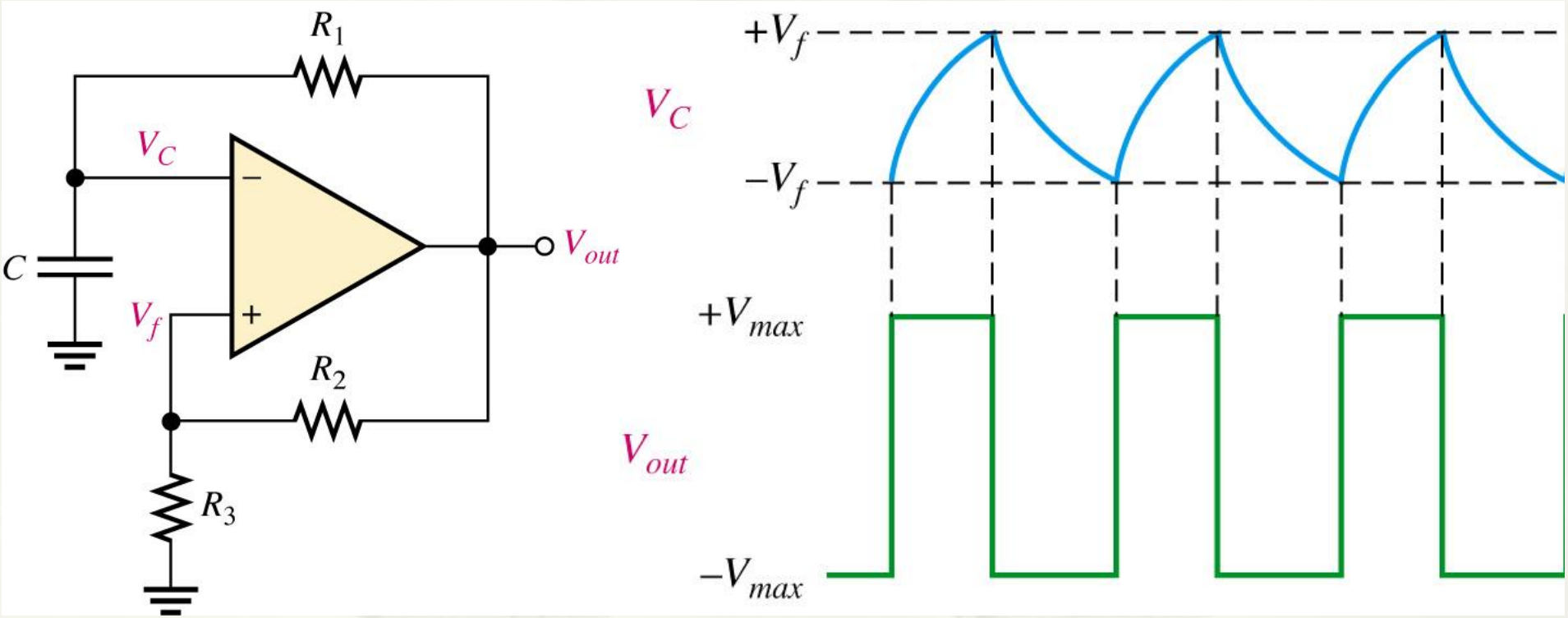
Solution

$$f \cong \frac{1}{4R_1C}\left(\frac{R_2}{R_3}\right) = \left(\frac{1}{4(10\,\text{k}\Omega)(0.01\,\mu\text{F})}\right)\left(\frac{33\,\text{k}\Omega}{10\,\text{k}\Omega}\right) = \mathbf{8.25\ kHz}$$

To make $f = 20$ kHz,

$$R_1 = \frac{1}{4fC}\left(\frac{R_2}{R_3}\right) = \left(\frac{1}{4(20\,\text{kHz})(0.01\,\mu\text{F})}\right)\left(\frac{33\,\text{k}\Omega}{10\,\text{k}\Omega}\right) = \mathbf{4.13\ k\Omega}$$

A Square-Wave Oscillator



Calculate the frequency of this square-wave oscillator.

KEY OBJECTIVES

1. Describe the basic operating principles for all oscillators
2. Explain the operation of feedback oscillators
3. Describe and analyze the operation of basic *RC* sinusoidal feedback oscillators
4. Describe and analyze the operation of basic relaxation oscillators

HOMEWORK

- **Chapter Checkup (all)**
  - Just do on your textbook
- **Questions (all)**
  - Odd-numbered in your mind, even-numbered on exercise book
- **Basic Problems**
  - 1, 3, 4, 6, 8
- **Basic-plus Problems**
  - 13, 14, 15
- **Complementary Problems**
  - Calculate the frequency of the square-wave oscillator.